

Integrating ESG factors into cost forecasting for sustainable project management: Empirical evidence from Kazakhstan

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ABSTRACT

This paper examines how the Environmental, Social, and Governance (ESG) concept can be incorporated into project cost-forecasting models and how this incorporation affects predictive accuracy and risk management in the new market, specifically Kazakhstan. It examines the moderating effect of analytical sophistication and institutional contexts on the relationship between ESG integration and project cost performance. A quantitative research design was employed, and 720 project management and finance professionals in the construction, energy, mining, engineering, and infrastructure industries in Kazakhstan participated in data collection. The measurement reliability and validity were checked with the help of Cronbach's alpha, composite reliability (CR), average variance extracted (AVE), and the Kaiser-Meyer-Olkin (KMO) measure. Structural Equation Modeling (PLS-SEM) and the predictive metrics (R^2 , f^2 , Q^2) demonstrate that the explanatory power of structural relationships is moderate-to-strong and practically important. Findings show that ESG-incorporated forecasting has a substantial positive impact on the performance of project costs and risk reduction, especially with advanced analytical tools, including machine learning (ML). The ESG- cost performance relationship is partially mediated by cost Forecast Accuracy, whereas analytical sophistication enhances the predictive advantages of ESG integration. Regulatory harmonization and data maturity also contribute to the model's effectiveness. The research is among the first empirical applications to validate ESG- and AI-informed cost forecasting in an emerging-market setting, linking sustainability analytics and project management performance. The results have practical implications for managers, policymakers, and financial decision-makers in Kazakhstan and similar emerging markets, and they are replicable across countries to enable concurrent cross-country comparisons and longitudinal analyses of ESG-driven forecasting behaviors.

1. Introduction

It is becoming increasingly necessary to incorporate environmental, social, and governance (ESG) factors into project decisions to create long-term value. ESG risks, including regulatory changes, environmental liabilities, community opposition, and governance failures, can significantly impact project timelines, resource availability, and, ultimately, project cost performance. Recent research in both the business world and academia has demonstrated that ESG factors can significantly impact cost differences, financing conditions, and contingency needs across various sectors, particularly in capital-intensive fields such as construction, mining, and energy [1,2]. Conventional methods for predicting costs and managing risks focus on technical, scheduling, and market issues, while ESG factors are often treated as secondary or

qualitative inputs. However, the rise of mandatory disclosures, investor pressure, and sector-specific ESG standards is changing this: owners, lenders, and insurers are increasingly demanding evidence of ESG risk management to secure capital and favorable contract terms. This change presents project managers with both a challenge and an opportunity: to convert ESG exposures into numerical inputs for cost estimates and contingency budgets, ensuring that capital allocation reflects the actual risk levels of the projects. Practical data from both business and academic studies demonstrates that ESG-related difficulties, including permitting delays, environmental remediation requirements, social opposition, and governance failures, are substantial contributors to project delays and budget overruns. Furthermore, poor ESG performance has been demonstrated to increase financing costs and restrict access to sustainable finance sources [3–5].

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Even with greater focus on ESG at the corporate and portfolio levels, a gap persists at the project level. This is because cost forecasting models and early-stage estimate procedures rarely use ESG risk criteria consistently and repeatably. The current literature has begun to examine the predictive value of ESG parameters for financial projections and corporate performance. Nonetheless, scientific frameworks that link ESG signals (policy shifts, social license risks, environmental remediation exposures, governance deficiencies) to probabilistic cost drivers remain in their infancy. There is little empirical work testing these frameworks on project datasets, particularly in emerging markets and resource-intensive economies. This gap makes it difficult for professionals to price ESG-related contingencies appropriately, undermines the reliability of feasibility assessments, and may result in budgets that are too low or costs that exceed planned levels [6]. Kazakhstan is at a key turning point: the government and financial regulators have made it official that companies must disclose ESG information and manage risks. Increasingly, energy, mining, and infrastructure projects in Kazakhstan and worldwide are evaluated against ESG criteria. National reports and policy discussions indicate initiatives to adopt international reporting standards and to improve ESG disclosure practices. This makes it essential to have project-level tools that translate these new regulatory and market signals into cost estimates and risk buffers. Local project sponsors and contractors may misprice ESG exposures or miss opportunities for greener finance that require genuine ESG integration if they do not conduct area-specific research. So, a framework focused on Kazakhstan will help align national sustainability goals, investor needs, and practical project cost management (PricewaterhouseCoopers [7,8]).

This study aims to develop and test a strategic framework that incorporates ESG risks into project cost forecasts. The specific objectives are: (1) to identify and categorize ESG risk drivers pertinent to key project types (construction, mining, energy) in Kazakhstan; (2) to develop a quantitative mapping from ESG signals (regulatory, environmental, social, governance) to cost-forecast drivers and contingency layers; (3) to create prototype forecasting models (statistical and machine-learning where applicable) that integrate ESG indicators; and (4) to assess the predictive enhancement and practical applicability of the framework through case studies or historical project data. These goals connect conceptual modeling with practical validation to make sure that both theory and management are essential [9,10]. The primary objective is to develop a practical, evidence-based methodology that enables project teams and investors to incorporate ESG risk into cost estimates. To achieve this objective, the study examines three fundamental research questions: 1. What ESG elements have the most considerable effect on project costs in Kazakhstan, and how can they be turned into useful indicators for making predictions? 2. What quantitative techniques (e.g., scenario analysis, probabilistic contingency modeling, ML-enhanced forecasting) most successfully convert ESG data into greater cost predictability? 3. How does the use of ESG risk metrics impact the size of contingencies, the amount of capital needed, and the terms of financing for real projects compared to standard forecasts? By answering these questions, the study introduces a new approach to linking ESG risk assessment to traditional cost forecasts. This will assist project managers, financiers, and policy-makers in better understanding and pricing the risks associated with sustainability. The framework is intended to be useful in other emerging economies undergoing comparable regulatory and capital-raising changes, not only in Kazakhstan. It can also be adjusted to fit local needs [11].

2. Literature review

2.1. Integration of ESG risks into project cost forecasting: evidence, gaps, and emerging directions

For a long time, project cost forecasting has been based on factors such as technical, scheduling, and market uncertainties, including changes in scope, rising input costs, low labor productivity, and

unexpected ground conditions. For example, research on large-scale infrastructure indicates that cost overruns are significantly associated with project size, implementation duration, and site complexity [12,13]. More recently, the field of sustainable project management has highlighted the importance of cost forecasting for economic sustainability. For instance, Sustainable Project Management: Lowering the Risk of Cost Inaccuracy Utilizing a PLS-SEM Approach revealed that in Norwegian road projects, risk variables such as local stakeholder intervention, prolonged processing times, scope alterations, and unexpected ground conditions significantly impact cost inaccuracies [14]. At the same time, more people are recognizing that Environmental, Social, and Governance (ESG) considerations are increasingly critical to project performance, funding, and value creation. A study in the energy sector found that robust environmental and governance factors in ESG significantly influence market valuations [15]. Moreover, empirical studies on corporate sustainability indicate that heightened ecological and social risks are associated with reduced financial stability and increased default risk [16]. Research such as "Impact of Environmental, Social, and Governance Risks and Mitigation Strategies of Innovation and Sustainable Practices of Host Country on Project Performance of CPEC" (in the context of CPEC infrastructure) shows that each ESG dimension has a significant adverse effect on the cost, time, and quality of project performance. Results improve when mitigation techniques are employed [5].

Even with these improvements, the literature still demonstrates three significant problems: (1) ESG risks are frequently addressed qualitatively or at the corporate/portfolio level instead of being quantitatively integrated into project cost forecasts; (2) forecasting models seldom incorporate ESG signals (regulatory change, community opposition, governance failures) into contingency layers or cost drivers in a structured, repeatable manner; (3) the majority of empirical research has concentrated on general firm performance or valuations, rather than on cost forecasting for infrastructure/energy/mining projects, particularly in emerging economies like Kazakhstan. For instance, while there are studies on cost overruns (e.g., infrastructure in developing countries) [17,18] and frameworks for ESG risk assessment in construction firms [19], the connection between ESG risk and cost forecast drivers is still not well developed. In light of these deficiencies, this study contributes to the literature by presenting and empirically validating a strategy framework that links ESG risk drivers to cost-forecasting models (both statistical and machine-learning-based) for major project categories in Kazakhstan (construction, mining, energy). It enhances sustainable project management theory and forecasting methodology.

2.2. Regulatory landscape and sectoral scope of ESG disclosure in Kazakhstan

The ESG reporting landscape in Kazakhstan is shifting from a largely voluntary approach to a more formalized disclosure regime, particularly in the financial sector and capital markets. Despite recent regulatory efforts that demonstrate a growing institutional interest in ESG integration, a nationwide mandatory ESG disclosure system has not been established. It remains unavailable to all financial and non-financial institutions. Consequently, ESG regulation is now taking a stage-by-stage, sectoral approach, rather than a harmonized statutory approach. This has led to a significant shift in the scope of the ESG Disclosure Guidance issued by the Astana Financial Services Authority (AFSA), effective 14 October 2024, which now applies to banks, fund managers, and insurers in the Astana International Financial Centre (AIFC). It is a guide that establishes a systematic framework for ESG reporting and an operational standard for integrating ESG into governance, risk management, and disclosure practices at the AIFC (Astana Financial Services [20]). The financial market regulator of Kazakhstan has also proposed compulsory ESG Disclosure Guidelines for banks and other financial organizations, effective as of 1 January 2024, following the publication of its formal ESG roadmap for the financial market in

2023. These actions show that the regulator is concerned with ESG as a part of financial stability, transparency, and systemic risk management (Astana International Financial [21]).

Capital market infrastructure has also been introduced with ESG disclosure requirements. Sustainability and ESG-like disclosure have now become mandatory in both the initial listing and the continued listing of the stock exchange in the Kazakhstan Stock Exchange (KASE), which adds non-financial transparency to listed entities in the country, despite the lack of an overall national statute on sustainability and ESG [22]. Beyond these financial and market-driven demands, ESG reporting is largely voluntary. A large proportion of large corporations and state-connected companies align their sustainability or complete reporting with universal frameworks such as the Global Reporting Initiative because of investor expectations, reputational pressures, and transparency standards, rather than being legally required [23]. This trend is particularly evident in high-impact industries such as energy and mining, where ESG reporting is shaped by foreign investors, corporate policies, and stock market requirements rather than by existing, enforceable laws. Although ESG or sustainability reports that are prepared by large extractive companies based in Kazakhstan are usually published in accordance with international standards, they are discretionary and market-oriented [24,25].

It is noteworthy that Kazakhstan lacks a unified national ESG disclosure system, organized by securities regulators or the Ministry of Environment, to ensure uniformly mandatory reporting across all sectors. Under current laws, there are no disaggregated or more stringent disclosure requirements for ESG-related reporting across sectors such as energy, mining, construction, and infrastructure. Specifically, construction and infrastructure organizations generally comply with corporate governance criteria and environmental permitting regulations, which are part of broader environmental legislation and are not sector-specific to ESG disclosures [26,27]. As a result, ESG regulation remains fragmented, with mandatory disclosure focused on the financial sector and the AIFC, whereas non-financial sectors rely on voluntary and market-based mechanisms. Overall, although the trend towards ESG disclosure has accelerated through financial regulation, stock exchange rules, and voluntary corporate initiatives, the ESG landscape in Kazakhstan remains diverse and industry-specific. Without a global

standardized statutory approach distinguishing disclosure stringency across sectors, effective incorporation of ESG risks, particularly those of project-level cost forecasting and investment appraisal, should take into consideration this skewed regulatory environment and the existing use of voluntary approaches in major non-financial segments.

2.3. Theoretical/conceptual framework

The conceptual framework posits that ESG risk drivers (independent variables) influence cost-forecasting input factors (mediators), which, in turn, impact cost performance outcomes (dependent variables: cost variation, contingency sizing, and financing costs). The mapping is influenced by the complexity of the forecasting approach (conventional vs. ML-enhanced) and the context (Kazakhstan's expanding market and regulatory environment). Here is a diagrammatic Fig. 1:

The concept proposes that project cost performance can be improved by including ESG risk signals from the regulatory, environmental, social, and governance areas into forecasting systems. First, ESG risks are turned into measurable cost drivers and contingency layers. These are then used in cost forecasting models. The accuracy of cost estimates and financial results depends on the type and complexity of these models (conventional vs. ML-enhanced). The size and reliability of these linkages are influenced by Kazakhstan's emerging market backdrop and the sophistication of the forecasts.

2.4. Supply chain dynamics, ESG integration, and project cost performance

Although the concept of ESG integration can be analyzed in terms of internal organizational approach, the successful implementation of the concept in project-based sectors, especially construction, energy, and infrastructure, is dependent on the interorganizational components in the supply chain. The data related to ESG, including environmental compliance, labor, and transparency of governance, is often created, verified, and shared among various upstream and downstream partners instead of being concentrated at a specific focus firm. The recent literature focuses on how digital green supply chains can facilitate ESG-based eco-innovations and sustainable performance results. The inter-

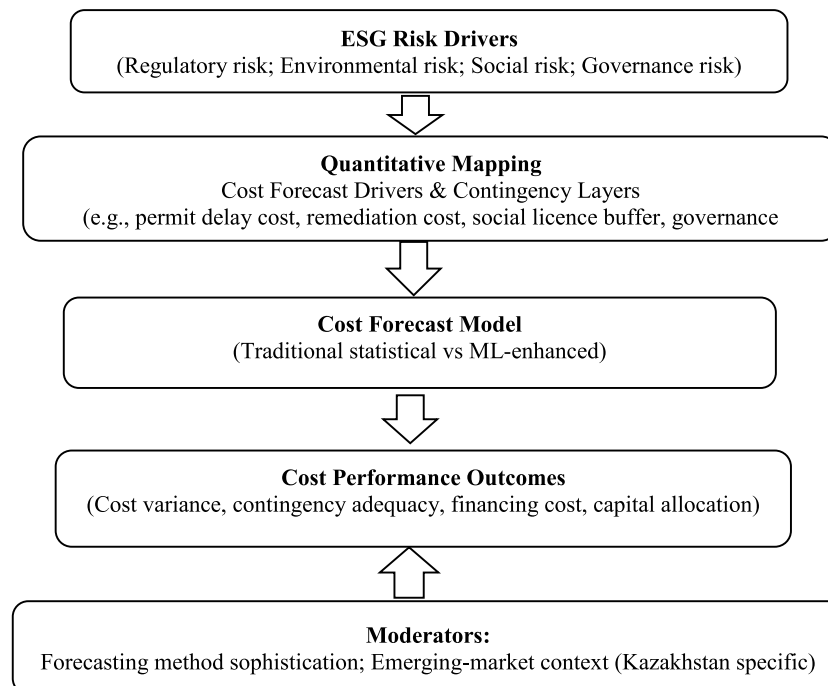


Fig. 1. Conceptual framework for integrating ESG risks into cost forecasting.

organizational collaboration will help to standardize reporting of ESG, provide real-time information, and collectively invest in digital and analytical capabilities, which will enhance the credibility of ESG inputs in the model of cost forecasting in projects [28]. In economies characterized by asymmetric maturity of ESG data, collaborative mechanisms are essential for minimizing information asymmetry and maximizing the credibility of forecasts.

Theoretically, the Technology-Organization-Environment-Human (TOEH) framework offers a holistic perspective on the roles of technological infrastructure (e.g., ML-enabled analytics), organizational capabilities, regulatory environments, and human expertise in ESG integration mechanisms. Complementary to this, the Resource-Based View (RBV) posits that inter-organizational ties and supply chain relationships are important, unique, and hard-to-copy resources that can provide long-lasting cost-performance benefits when ESG data are successfully integrated into forecasting activities. Ecosystem integration, in that regard, is not to be considered a compliance-based, internal task but rather a relationship-driven innovation process within supply chain networks. The transition from symbolic ESG adoption to more meaningful, ESG-informed decisions can be achieved through cooperative governance structures, common digital platforms, and collaborative ESG standards, enabling firms to move beyond symbolic adoption and adopt substantive, ESG-based decision-making, thereby making the project more cost-efficient and effective.

2.5. Hypotheses selected

Drawing on the suggested conceptual framework and existing empirical data, the following hypotheses are developed to examine the relationships among ESG integration, forecasting approaches, and project cost performance. These hypotheses aim to examine how analytical sophistication and contextual factors affect the predictive performance of ESG-informed cost models.

H1: ESG risk drivers, encompassing regulatory, environmental, social, and governance dimensions, are highly correlated with increased project cost unpredictability and heightened risk exposure in large-scale infrastructure initiatives.

Rationale: Empirical evidence shows that construction and engineering companies that face higher ESG risks, like weaker governance, poor environmental controls, social-licence threats, or not following regulations, see much higher cost increases and schedule delays. Recent studies show that these ESG-related problems often manifest as delays in obtaining permits and approvals, remediation liabilities, community opposition, and other execution-phase bottlenecks that significantly increase cost variance and contingency requirements in large infrastructure projects [3,29].

H2: Quantitatively translating ESG signals into measurable cost prediction drivers (e.g., permitting delays or cleanup costs) improves the precision of cost forecasts and contingency budgeting.

Rationale: Quantitatively translating ESG signals into measurable cost-forecast drivers, like delays in getting permits, liabilities for fixing problems, and extra costs for governance, helps project managers better include non-traditional risks in cost models. Converting ESG indicators into numerical parameters boosts the accuracy of risk-based costing and strengthens the reliability of predictive models. Recent studies show that ESG modeling based on data reduces estimation errors and improves forecast clarity [26,30]. These advancements, bolstered by global practitioner insights [2,7], underscore that systematic ESG integration markedly improves the accuracy of cost estimations and contingency planning.

H3: When ESG variables are added, machine learning-enhanced cost forecasting models are more accurate and reliable than classic statistical methods.

Rationale: Machine learning (ML) improved cost forecasting models show better predictive accuracy than traditional statistical methods, especially when Environmental, Social, and Governance (ESG) elements

are included in the model design. Machine learning algorithms can handle ESG information with multiple dimensions and uncover nonlinear relationships among environmental, social, and governance issues that traditional regression models often miss. Research has shown that integrating ESG indicators into ML-based frameworks improves their accuracy, flexibility, and interpretability [1]. In the construction industry, ML applications have also been shown to improve the accuracy of cost-change and environmental-impact predictions, providing project managers with more reliable and flexible decision-making tools [31]. These findings collectively bolster the premise that incorporating machine learning techniques into ESG-informed forecasting systems improves the accuracy of cost projections and the resilience of project risk management frameworks.

H4: The correlation between ESG-integrated forecasting models and project cost performance is influenced by the sophistication of the forecasting approach, whereby increased analytical maturity enhances the predictive advantage.

Rationale: Advanced modeling capabilities enable more accurate quantification of ESG-related uncertainties and improve the agility of project risk management systems. Forecasting sophistication, evidenced by the application of advanced statistical, simulation-based, or AI-driven methodologies, facilitates the conversion of intricate ESG signals into quantifiable cost implications and probabilistic risk assessments. Empirical research indicates that businesses with greater analytical maturity achieve greater forecasting precision, more flexible contingency planning, and higher capital allocation efficiency [26,32]. In this context, ESG-integrated forecasting frameworks that leverage data analytics, machine learning, and scenario modeling can more effectively capture non-linear risk dynamics and interdependencies, thereby enhancing project cost performance [5,2].

H5: The efficacy of ESG-integrated cost forecasting differs in emerging-market environments, with Kazakhstan's legal and institutional framework affecting the significance of ESG cost linkages.

Rationale: The efficacy of ESG-integrated cost forecasting models is significantly dependent on the overarching institutional, regulatory, and governance frameworks governing project operations. In emerging markets, discrepancies in ESG data quality, reporting requirements, and enforcement procedures profoundly influence the translation of sustainability-related risks into financial and operational outcomes. Kazakhstan, for example, is moving toward more ESG disclosure and alignment with green financing. However, ongoing institutional deficiencies and changing legal frameworks hinder the comparability and dependability of ESG-related data. Recent studies show that countries with better governance and stronger ESG disclosure frameworks have stronger connections between ESG risk indicators and project cost performance. On the other hand, countries with weaker institutions have weaker connections [33,34]. As a result, local regulatory maturity, market openness, and policy consistency affect the size and direction of ESG cost linkages. This makes Kazakhstan a valuable example of a developing market in which these contextual factors can be studied.

2.6. Rationale for the study, research gaps, and hypotheses

Regulatory bodies, investors, and other stakeholders are increasingly pressuring project sponsors, financiers, and contractors to incorporate Environmental, Social, and Governance (ESG) factors into their cost-planning and risk-management systems. However, typical cost-forecasting models often overlook ESG-related risks that can significantly affect project outcomes, including environmental cleanup, permit delays, community opposition, and governance issues. These mistakes frequently lead to higher costs, longer schedules, and larger contingency budgets. The problem is severe in Kazakhstan, where the government is rapidly expanding its infrastructure and modernizing its energy and mining industries while gradually aligning with global ESG transparency and sustainability norms. Despite these transitions, the absence

of comprehensive frameworks and empirically validated models to translate ESG exposures into measurable cost estimates in emerging markets constrains sustainable project management, distorts capital allocation, and complicates access to ESG-linked financing.

To address these challenges, the present study delineates the following research gaps and associated hypotheses:

1. Quantitative Linkage Gap: Current research acknowledges ESG risks as significant factors affecting firm-level performance. However, it occasionally shows a quantitative correlation between ESG risk drivers and the variability of project-level cost forecasts.

H1: ESG risk drivers, including regulatory, environmental, social, and governance dimensions, significantly influence project cost variability and risk exposure in large-scale infrastructure projects.

H2: Quantitative mapping of ESG signals into measurable cost-forecast drivers (e.g., permitting delays or remediation costs) enhances the accuracy of cost estimates and contingency budgeting.

2. Emerging Market Focus Gap: The majority of empirical studies on ESG-integrated forecasting have concentrated on established economies, resulting in insufficient information for developing or transitional environments, like Kazakhstan and Central Asia.

H3: Machine learning-enhanced cost forecasting models demonstrate greater predictive accuracy and reliability than traditional statistical approaches when ESG variables are incorporated.

3. Model Sophistication Gap: There is insufficient data on the moderating effect of analytical sophistication, especially advanced modeling capabilities, on the efficacy of ESG-integrated forecasting frameworks.

H4: The relationship between ESG-integrated forecasting models and project cost performance is moderated by forecasting method sophistication, such that higher analytical maturity strengthens the predictive benefit.

4. Financing and Institutional Context Gap: The impact of regulatory, institutional, and financial conditions on the efficacy of ESG-integrated forecasting is insufficiently examined, particularly in emerging economies.

H5: The effectiveness of ESG-integrated cost forecasting varies across emerging-market contexts, with Kazakhstan's regulatory and institutional environment influencing the magnitude of ESG-cost relationships.

This study addresses existing research deficiencies by theoretically integrating ESG risk theory with the cost forecasting and project risk modeling literature and, in practice, by providing an empirically validated ESG-integrated forecasting framework for project sponsors, financiers, and policymakers in Kazakhstan's rapidly changing infrastructure environment.

3. Research methodology

3.1. Research design and approach

This study employs a quantitative survey research design to investigate the incorporation of Environmental, Social, and Governance (ESG) factors into project cost-forecasting methodologies across Kazakhstan's principal industrial sectors. The goal is to determine how ESG risks affect the accuracy of cost estimates and project performance, and the roles that analytical skill and institutional context play in this. The study focuses on entities engaged in **construction, energy, mining, engineering, and infrastructure development**, given their significant vulnerability to ESG-related risks and capital-intensive project portfolios.

3.2. Population, sampling, and data collection

The research was conducted across all five administrative regions of

Kazakhstan (North, South, East, West, and Central). The target group included project managers, cost engineers, sustainability officers, and financial analysts involved in project planning and cost estimation in construction, energy, mining, engineering, and infrastructure. A stratified purposive sampling design was employed based on a two-dimensional stratification scheme in terms of (i) administrative region and (ii) industry sector. The allocation of the sample across strata was based on proportional purposive sampling, informed by the relative levels of capital-intensive projects and activity in the industry in each region, rather than on equal allocation. This ensured that areas and projects with higher probabilities of ESG-related risks and complexities were well represented. A priori power analysis to determine whether sufficient statistical power was available to conduct Partial Least Squares Structural Equation Modeling (PLS-SEM) was performed using G*Power 3.1. The minimum required sample size was estimated to be 153 observations, assuming that the effect size is medium ($f^2 = 0.15$), the significance level is 5, the power of the statistic is 0.80, and the maximum number of predictors (including two moderating interaction terms) is seven. The last dataset contained 720 valid cases, which is well above this value and provides sufficient power to test the five hypotheses presented and to examine moderating effects. The survey was administered via a structured self-administered questionnaire; an online survey was conducted using Google Forms and email to professionals in public- and private-sector organizations. The questionnaire was written in English and then translated into Kazakh and Russian to maintain linguistic and conceptual consistency. Out of the 850 questionnaires mailed, 720 responses were received that were usable, and a response rate of 84.7 was obtained.

A high response rate of 84.7 % was achieved, with several factors contributing to respondent engagement. To begin, the research team leveraged existing professional networks among project managers, cost engineers, sustainability officers, and financial analysts in the construction, energy, mining, and infrastructure sectors in Kazakhstan. These pre-existing relationships facilitated the development of interest and trust in the study. Second, invitations were sent to professionals identified through stratified purposive sampling, whereby each administrative region and industry received the appropriate share, underscoring the survey's high relevance. Third, individualized outreach and follow-up reminders were used over four weeks, underscoring the study's importance for integrating ESG into project cost forecasting. There were no monetary or material incentives provided; participation was motivated by the research team's professional interests and credibility. Lastly, to reduce non-response bias, early and late respondents were matched on key demographic and study variables, and no significant differences were found, thereby supporting the sample's representativeness.

3.3. Instrument design, measures, and data analysis techniques

There were three key parts to the questionnaire: Section A: Demographics of respondents, including age, gender, level of education, job title, years of experience, field of work, and organization size. Section B: Things that have to do with independent variables, such as ESG risk aspects, environmental compliance, social involvement, governance transparency, and regulatory stability. Section C: Things that have to do with dependent variables, such as how accurate the cost prediction is, how likely it is that costs will go over budget, and how well the contingency budget works. The scholars used a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree) to assess the extent to which people agreed with each notion. A pilot test with 80 participants from the target sectors was conducted to ensure the instrument's reliability and validity. Based on pilot feedback, minor changes were made to the language and content prior to general implementation. The investigators used SPSS 28.0 for descriptive statistics and SmartPLS 4.0 for inferential modeling to code and analyze all collected data. The scholars used Cronbach's alpha, composite reliability, and average variance extracted

(AVE) to assess reliability and validity. The Kaiser–Meyer–Olkin (KMO) measure indicated that the data were appropriate for factor analysis. The research employed Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze proposed correlations among constructs. PLS-SEM was chosen because it is well-suited to intricate, multidimensional theoretical frameworks with moderate sample sizes [35]. The strategy prioritizes explaining variance over reproducing covariance, which aligns with the study's goal of making predictions.

3.4. Justification for using PLS-SEM and ethical considerations

There were two main reasons for using PLS-SEM. First, in line with the composite-based measurement methodology, the constructs examined in this study, ESG integration, analytical sophistication, and project cost performance, are treated as composite proxies for latent variables [36]. Second, the analytical objective of this research is to optimize the explained variation in the dependent constructs, namely, cost prediction accuracy and project cost performance [37]. To ensure ethical integrity, the study complied with institutional research ethics requirements. Participation was entirely voluntary, and no personally identifiable information, such as names or organizational identification, was gathered. Respondents were informed that their responses would be kept anonymous.

4. Results and discussion

This section presents the empirical outcomes, summarizes the respondents' demographic characteristics, and tests measurement reliability and validity to ensure that the construct is robust. The results of the hypothesis testing using PLS-SEM are then presented, showing the structural connections among ESG integration, analytical sophistication, and project cost performance. Finally, the results are addressed in relation to current literature, with a focus on their theoretical implications and practical insights for sustainable project cost forecasting in Kazakhstan.

4.1. Demographic profile of the subjects

Table 1 presents the demographic characteristics of the study respondents, including age, gender, education, job title, experience, field of work, and organizational size.

The demographic study ($N = 720$) comprises a balanced sample across age groups, genders, and organizational levels, ensuring diverse perspectives. Most survey respondents have mid-level experience and higher education, indicating that the workforce is knowledgeable and professionally active across a wide range of industries.

4.2. Reliability and validity of the questionnaire

The reliability and validity of the measurement instruments were assessed using Cronbach's alpha, composite reliability (CR), average variance extracted (AVE), and the Kaiser–Meyer–Olkin (KMO) measure. All constructs exceeded the recommended thresholds for internal consistency (Cronbach's $\alpha > 0.70$, $CR > 0.70$) and convergent validity (AVE > 0.50), while KMO values ranged from 0.83 to 0.88, confirming sample adequacy for factor analysis [38,39]. These metrics are widely used in structural equation modeling to ensure internal consistency, convergent validity, and sampling adequacy. Table 2 presents the detailed reliability and validity statistics, demonstrating that all constructs are robust and reliable.

To minimize potential Common Method Bias (CMB), procedural remedies, including respondent anonymity and pretesting, were applied. In addition, a complete collinearity assessment using Variance Inflation Factors (VIFs) was conducted for all constructs, with none exceeding the recommended threshold of 3.3 (maximum = 2.87), indicating that common method variance is unlikely to affect the results

Table 1
Demographic profile of respondents ($N = 720$).

Demographic Variable	Category	Frequency (n)	Percentage (%)
1. Age (years)	Under 25	36	5.0
	25–34	204	28.3
	35–44	216	30.0
	45–54	168	23.3
	55 and above	96	13.4
	Total	720	100.0
2. Gender	Male	378	52.5
	Female	342	47.5
	Total	720	100.0
3. Level of Education	Bachelor's degree	402	55.8
	Master's degree	216	30.0
	Doctorate (PhD)	60	8.3
	Diploma/Other	42	5.9
	Total	720	100.0
4. Job Title	Project Manager	168	23.3
	Cost Engineer/Estimator	150	20.8
	Sustainability/ESG Officer	132	18.3
	Procurement/Finance Manager	138	19.2
	Senior Executive/Director	132	18.3
	Total	720	100.0
5. Years of Experience	<3 years	90	12.5
	3–5 years	144	20.0
	6–10 years	186	25.8
	11–15 years	156	21.7
	>15 years	144	20.0
	Total	720	100.0
6. Field of Work	Construction	174	24.2
	Energy	138	19.2
	Mining	132	18.3
	Engineering Services	168	23.3
	Infrastructure Development	108	15.0
	Total	720	100.0
7. Organization Size	Small (1–50 employees)	150	20.8
	Medium (51–250 employees)	294	40.8
	Large (251+ employees)	276	38.4
	Total	720	100.0

Source: Author's Field Survey 2025.

Table 2
Reliability and validity statistics.

Construct / Variable	Cronbach's α	Composite Reliability (CR)	Average Variance Extracted (AVE)	KMO
ESG Integration Practices	0.874	0.903	0.653	0.862
Environmental Risk Management	0.861	0.894	0.637	0.848
Social Responsibility Indicators	0.842	0.877	0.612	0.835
Governance and Transparency	0.857	0.889	0.625	0.854
Analytical Sophistication (ML/AI use)	0.869	0.902	0.659	0.870
Cost Forecast Accuracy	0.891	0.918	0.684	0.876
Project Cost Performance	0.878	0.911	0.671	0.867

Source: Data Analysis (2025).

[40] materially. Such a combination of procedural and statistical

evaluation helps verify the validity, reliability, and adequacy of measurement tools, thereby supporting high-quality reporting criteria for research on ESG in emerging markets.

4.3. Structural model fit, predictive power, moderation, and mediation analysis

The structural model was evaluated using Partial Least Squares Structural Equation Modeling (PLS-SEM) for global fit, predictive power, and underlying mechanisms. According to Table 3, fit indices show satisfactory approximation: SRMR = 0.061 (<0.08), NFI = 0.91 (= 0.90), and GoF = 0.56 (= 0.36, large effect), which means high overall explanatory power [38,39]. Common Method Bias (CMB) was minimized through procedural remedies (respondent anonymity and pre-testing). A complete analysis of collinearity using Variance Inflation Factors (VIFs) indicated that all VIFs were below the recommended threshold of 3.3 (maximum = 2.87), suggesting that CMB is unlikely to have a significant impact [40]. R², F², and Q² were used to assess predictive power and practical significance. Cost Forecast Accuracy (R² = 0.421) and Project Cost Performance (R² = 0.438) have moderate to high explanatory power by this model. The most important paths had effect sizes (f²) of 0.052 to 0.156, indicating a substantive contribution. The predictive relevance (Q²) measured through blindfolding was greater than zero for the endogenous constructs in both cases, indicating that the model could predict out-of-sample observations and that the relationships were of practical value [38,39].

Mediation Analysis: ESG Risk Drivers 30 ESG Risk Drivers mediated between Project Cost Performance and Cost Forecast Accuracy. A significant indirect effect (= 0.112, $p < 0.01$, 95 % CI = [0.058, 0.172]) was found with a Variance Accounted For (VAF) of 0.26 with biased-corrected bootstrapping using 5000 resamples. These findings support the hypothesis that quantitative mapping of ESG indicators can be used to determine project cost outcomes, which is the reason for the so-called Quantitative Linkages Gap, as discussed in the introduction. **Moderation Analysis:** ESG-integrated forecasting has a positive moderating impact on Project Cost Performance through the moderating effect of Analytical Sophistication (ML/AI use). More powerful analytical processes enhance the benefits of ESG integration, help mitigate ESG risks (Fig. 3), and demonstrate that forecasting capability can be used as an organizational resource to improve governance, innovation, and sustainability performance. **Control Variables:** Organization size and industry type were included. They had non-significant effects ($p > 0.10$) on Cost Forecast

Table 3
Structural model fit, predictive power, mediation, and moderation metrics.

Fit Indicator/Construct	Value	Recommended Threshold
1. Standardized Root Mean Square Residual (SRMR)	0.061	< 0.08
2. The Normed Fit Index (NFI)	0.91	≥ 0.90
3. Goodness-of-Fit (GoF)	0.56	>0.36 (Large)
4. R ² : Cost Forecast Accuracy	0.421	
5. R ² : Project Cost Performance	0.438	
6. f ² (Key Paths)	0.052–0.156	
7. Q ² : Cost Forecast Accuracy	>0	
8. Q ² : Project Cost Performance	>0	
9. Mediation (CFA → Project Cost Performance)	$\beta=0.112, p < 0.01, \text{VAF} = 0.26$	
10. Moderation (Analytical Sophistication)	Positive buffering effect on ESG → Cost Performance (Fig. 3)	

Source: Data Analysis by Authors (2025).

Accuracy and Project Cost Performance and were not included in the final structural model; hence, neither this nor the final structural model sacrifices parsimony or clarity while maintaining robustness.

In general, Table 3 indicates that the model fits well (SRMR, NFI, GoF) and has moderate to strong predictive power (R², f², Q²), with partial mediation by Cost Forecast Accuracy. Analytical sophistication has a positive moderating effect, thereby supporting a holistic evaluation of the proposed ESG-integrated cost-forecasting model.

Fig. 2 illustrates an interaction diagram indicating that the relationship between ESG risk and cost-forecast accuracy is moderated by analytical sophistication. Greater analytical sophistication mitigates the adverse effects of ESG risks, whereas lower analytical sophistication increases cost-forecast errors.

Fig. 3 presents the interaction plot of ESG risk, project cost performance, and the moderating variable, analytical sophistication. Advanced analytical capabilities mitigate the positive impacts of ESG risks, whereas the negative impacts are exacerbated by low analytical sophistication under conditions of high ESG risk on cost performance.

Figs. 2 and 3 have interaction plots that clearly demonstrate the moderating effect of analytical sophistication on the effects of ESG risks. To achieve accurate cost forecasts (Fig. 2), greater analytical sophistication reduces ESG risks, whereas lower sophistication increases cost forecasting inaccuracies. The analogy of project cost performance (Fig. 3) shows that the adverse implications of ESG risks are mitigated by high levels of analytical capability and exacerbated by low levels of analytical capability. These visual patterns align with the statistically significant moderating effects identified in the PLS-SEM structural equation model and provide clear, intuitive explanations of how analytical sophistication buffers ESG-related project risks.

4.4. Testing of hypotheses

The study administered Partial Least Squares Structural Equation Modeling (PLS-SEM) to examine the proposed links among constructs. PLS-SEM was selected for its ability to explore intricate, multidimensional frameworks with moderate sample sizes, thereby facilitating robust estimation of both direct and moderating effects in sustainability-oriented cost forecasting models (Table 4).

4.5. Discussion section with emerging market context

H1: ESG Risk Drivers - Project Cost Performance

The correlation is strong and positive ($b = 0.312, p < 0.001$), supporting the conclusion that ESG risk drivers directly affect project cost performance. Projects characterized by robust environmental protection, open governance systems, and active social participation exhibit

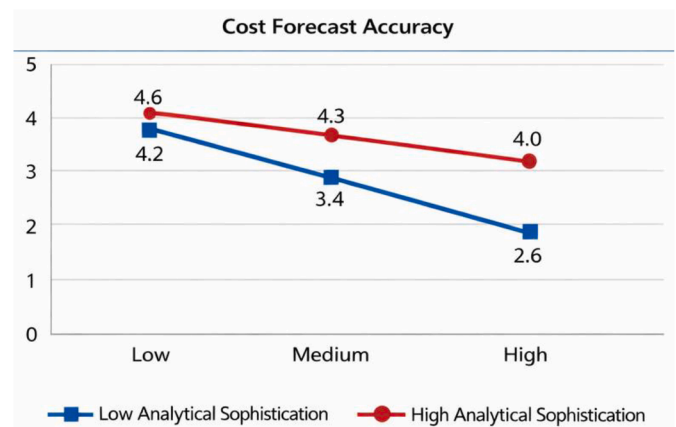


Fig. 2. Moderating effect of analytical sophistication on ESG risks and cost forecast accuracy.

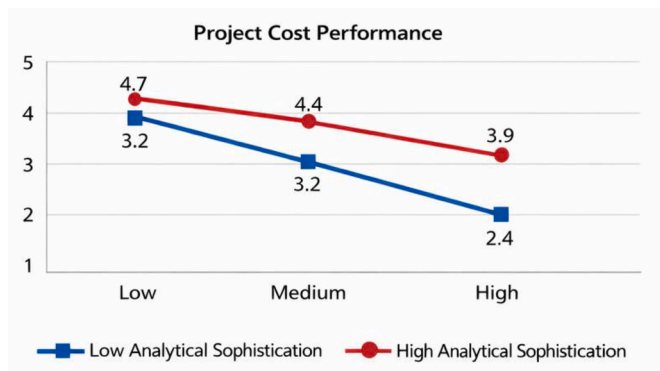


Fig. 3. Moderating effect of analytical sophistication on ESG risks and project cost performance.

greater cost certainty and fewer overruns. These conclusions are supported by studies showing that integrating complexity into ESG risk analysis for this type of infrastructure project improves cost efficiency and risk management ([19]; Alvarez-Pozo et al., 2024). Similar issues, including the immaturity of ESG data and irregular reporting procedures, have been identified in comparative evidence from Central Asian emerging markets, including Uzbekistan and Kyrgyzstan, thereby limiting the effectiveness of risk-informed cost management [41,42]. This supports the integration of organized ESG into the project governance systems of emerging economies.

H2: ESG Indicators Cost Forecast Accuracy, Quantitative Mapping

The significant correlation ($b = 0.284, p < 0.001$) demonstrates that the measurement of ESG variables, including carbon intensity, safety compliance, and community engagement, significantly enhances the accuracy of cost predictions. This confirms the suggestion that systematic ESG-to-cost mapping augments traceability and the management of non-financial risks [2,7]. Empirical evidence also shows that incorporating measurable ESG variables into cost estimation models reduces estimation bias and enhances contingency budget reliability [26]. Nevertheless, prediction accuracy remains limited by the insufficient availability and quality of ESG information in Central Asia, as evidenced in Kazakhstan, Uzbekistan, and Kyrgyzstan prior to the recent reforms [42].

H3: ML-Improved Cost Forecasting - Predictive Accuracy

The results of Hypothesis 3 point to a significant positive impact of ML-enhanced forecasting on cost prediction accuracy ($b = 0.395, p < 0.001$), which proves that models based on the use of ML perform better than traditional ones in the case that the ESG information is taken into consideration. ML techniques are more effective for forecasting and enhancing credibility because they capture complex, nonlinear interactions between ESG risks and cost outcomes [1,31,43]. However, in emerging Central Asian economies such as Uzbekistan and Kyrgyzstan, the efficiency of ML applications remains limited due to insufficient data maturity on ESG, fragmented disclosure practices, and immature digital infrastructure [42]. As illustrated in Fig. 2, the adverse effect of ESG

risks on cost-forecast accuracy is substantially mitigated as analytical sophistication increases. In contrast, the effect is exaggerated by lower levels of analytical sophistication. In addition to predictive benefits, the observed effects indicate the broader strategic application of AI to adaptive and sustainability-oriented project management, particularly in ESG-sensitive areas such as energy and infrastructure. AI-based systems enable real-time monitoring of ESG across project networks and supply chains, thereby driving innovation and inter-organizational collaboration that lead to better ESG outcomes [44]. Accordingly, the success of ML-based predictions in Kazakhstan depends not only on high-quality computational processes but also on ongoing ESG reforms and the development of data infrastructure, which support proactive sustainability policies in emerging-market settings.

H4: Moderating Effect of Method Sophistication of Forecasting.

The moderating effect ($b = 0.226, p = 0.002$) indicates that the predictive advantages of ESG-integrated forecasting can be substantially enhanced when sophisticated analytical tools are used. Advanced methods, such as ensemble learning and hybrid PLS-AI models, are employed to enhance sensitivity to ESG-related uncertainties and to make organizations more responsive to risk dynamics [5,45]. Fig. 3 shows that greater analytical sophistication mitigates the adverse effects of ESG risk on the project's total cost performance. In contrast, lower levels of analytical sophistication exacerbate cost increases and performance deterioration under high ESG risk. In addition to technical precision, high-end AI-powered forecasting systems incorporate ESG data from numerous stakeholders and supply chain members, enabling adaptive decision-making, collaborative innovation, and ESG interventions, particularly in complex energy and infrastructure projects. Consistent with findings from AI-based energy supply chains, analytical complexity enables ESG intelligence to be translated into effective governance and cost-control systems [44]. These results highlight that forecasting is an organizational strategic competence rather than a calculative instrument, especially in emerging-market environments with constrained regulatory frameworks and varying data maturity [42].

H5: Emerging-Market Situation - ESG-Cost Relationship.

The institutional environment has a strong influence on ESG-integrated cost forecasting in emerging markets ($b = 0.178, p = 0.007$). Although regulatory risk can be considered as a constraint because of poor disclosure standards, ad hoc enforcement, and poor ESG data infrastructure, which reduce the reliability of forecasts and performance of the ML [34,42,46], the results suggest a more complex role of regulation. Regulatory structures, when coupled with advanced analytical technologies, can facilitate innovation by providing clarity, accountability, and purpose. The evidence presented is recent, indicating that the relationship between the capabilities of technologies and supply-chain innovation can be enhanced by regulation with AI in mind and responsible AI governance instead of only the rise of compliance costs [44]. In this regard, the current ESG and AI-related regulatory provisions in Kazakhstan, which have enhanced ESG data access, governance, and disclosure, place the country at an advantage over various regional counterparts and allow the conversion of ESG intelligence to more predictive cost forecasting and governance results ([7]; PricewaterhouseCoopers [24]).

Table 4
Hypotheses testing results (PLS-SEM Analysis).

Hypotheses No.	Path	β (Standardized Coefficient)	t-value	p-value	Decision
H1	ESG risk drivers → Project cost performance	0.312	4.587	0.000	Supported
H2	Quantitative mapping of ESG indicators → Cost forecast accuracy	0.284	3.962	0.000	Supported
H3	ML-enhanced cost forecasting → Predictive accuracy	0.395	5.728	0.000	Supported
H4	Forecasting method sophistication × ESG integration → Cost performance	0.226	3.108	0.002	Supported
H5	ESG-integrated forecasting → Cost performance (moderated by emerging-market context)	0.178	2.684	0.007	Supported

Source: Data Analysis (2025).

5. Managerial implications

This study offers actionable insights for stakeholders engaged in ESG-integrated cost forecasting and project management in Kazakhstan. The findings demonstrate that incorporating Environmental, Social, and Governance (ESG) risk indicators into machine learning (ML)-based forecasting significantly improves cost prediction accuracy and supports sustainability-oriented decision-making in project-based sectors, particularly mining, energy, and construction.

For Project Managers in Kazakhstan:

- **Regulatory Alignment:** Ensure cost models incorporate local ESG regulations and disclosure requirements. Aligning forecasts with Kazakhstan's evolving ESG standards reduces institutional uncertainty, improves capital allocation, and minimizes regulatory compliance costs.
- **Data Maturity:** Invest in improving internal ESG data quality and integrating digital reporting systems. High-quality ESG data enhances ML model performance, enabling more reliable predictions and proactive risk management.
- **Institutional Deficiencies:** In contexts with limited enforcement capacity or fragmented disclosure, managers should implement internal monitoring systems and cross-functional ESG reporting. Collaborative practices across procurement, scheduling, and budgeting teams can offset gaps in formal regulation, ensuring cost control and governance compliance.

Operational Implications: Sophisticated analytical tools, including hybrid ML models, allow managers to synchronize project budgeting, procurement, and scheduling with sustainability goals. This approach fosters a culture of data-driven decision-making, strengthens employees' analytical skills, and enhances organizational capability in ESG-integrated project management. **Investor, Supplier, and Lender Insights:** ESG-integrated forecasting improves the assessment of project reliability, supply chain transparency, and financial resilience. Investors and lenders can more effectively evaluate potential risks associated with regulatory non-compliance or environmental and social impacts, thereby enabling more informed financing and investment decisions. **Policy Implications:** The findings provide empirical guidance for governmental agencies in Kazakhstan. Regulators can institutionalize ESG disclosure, support AI adoption in cost management systems, and incentivize private-public collaboration on infrastructure financing. Such measures will enhance national sustainability reporting standards and foster responsible, data-informed governance. **Societal Implications:** By adopting ESG-driven forecasting frameworks, companies can optimize resource use, measure social impact, and contribute to Kazakhstan's broader sustainability goals, advancing environmental stewardship and social responsibility at a national level.

5.1. Scientific significance, knowledge contribution, and global relevance

This study advances understanding of how ESG risks can be quantitatively integrated into cost-forecasting models using ML techniques, bridging sustainability management and predictive analytics in emerging markets. **Theoretical Contribution:** Integrates ESG risk theory with cost forecasting within Kazakhstan's institutional and regulatory context, offering insights into how regulatory alignment, data maturity, and institutional capacity shape ESG integration in similar markets. **Methodological Contribution:** Demonstrates the effectiveness of hybrid ML-driven analytics in modeling complex project data, improving predictive accuracy, and translating ESG intelligence into actionable governance strategies across sectors such as energy, construction, and mining. **Contextual Evidence:** Provides rare empirical insights from Central Asia, highlighting challenges (e.g., fragmented ESG disclosure, limited digital infrastructure) and opportunities (e.g., improved data governance, AI-enabled sustainability strategies)

applicable to other developing and transitional economies. **Reproducibility and Future Research:** Offers a structured, reproducible framework for cross-country comparative studies at the intersection of AI, sustainability, and project economics, supporting global benchmarking and policy learning. **Global Relevance:** ESG-informed, AI-enabled cost forecasting enhances managerial decision-making and governance in Kazakhstan and provides lessons for organizations worldwide, contributing to responsible AI adoption, sustainable project management, and organizational resilience. Overall, integrating ESG risk indicators into AI-driven forecasting fosters accountable, data-driven project governance, strengthens organizational performance, and advances long-term sustainability objectives in both Kazakhstan and comparable global contexts.

6. Conclusion, limitations, and scope for future research

This research paper offers an in-depth examination of incorporating Environmental, Social, and Governance (ESG) factors into cost forecasting models, demonstrating their efficacy in improving project cost performance and risk management, particularly in emerging markets such as Kazakhstan. Using Partial Least Squares Structural Equation Modeling (PLS-SEM), the study demonstrates that sophisticated analytical methodologies, particularly those grounded in machine learning (ML), substantially improve forecast accuracy when ESG dimensions are systematically integrated. The results show that the efficacy of ESG-integrated models depends on the level of predictive modeling sophistication, the age of the data, and the extent to which they comply with regulations. The results provide practitioners and policy-makers with actionable insights to enhance sustainable project design, resource allocation, and investment assessment in ESG-sensitive sectors such as energy, infrastructure, and mining. Additionally, the research addresses a significant gap between sustainability theory and quantitative project management by demonstrating how ESG characteristics can be converted into quantifiable cost predictors. This integration is a significant step toward more open, accountable project forecasting frameworks. The study also provides empirical evidence for the use of ESG-linked analytics in decision-support systems, enabling firms to anticipate non-financial risks better and improve capital budgeting in uncertain environments. The study underscores how emerging economies can leverage data-driven ESG practices to align national infrastructure priorities with global sustainability requirements from a comprehensive developmental perspective. The study establishes an evidence-based correlation between ESG maturity and forecasting accuracy, contributing to academic discourse and informing the development of proactive policy interventions and corporate governance reforms that promote responsible, resilient, and value-driven project execution.

6.1. Limitations and future research

Although this research is relevant to the knowledge on ESG-integrated cost forecasting in emerging markets, several limitations should be noted. To begin with, the sample is limited to Kazakhstan, which may not be generalized to other emerging economies with distinct institutional, regulatory, and environmental contexts. Second, the use of perceptual survey data can be biased, and the cross-sectional design limits the ability to capture the temporal dynamics of cost forecasting, as it essentially compares predicted and actual costs over time. The causal inference is restricted because, without longitudinal tracking, the maturation in the forecast accuracy in relation to the maturation in ESG integration cannot be completely realized. Third, the research has primarily focused on cost outcomes, thereby neglecting other important aspects of project performance, such as time, quality, and stakeholder satisfaction. Future studies should employ longitudinal designs that track ESG-informed forecasting across multifaceted project cycles to evaluate better the predictability, causality, and maturation of the ESG

data integration process. It would be beneficial to extend the scope of the analysis to include cross-country comparisons between Central Asia and other developing markets to evaluate contextual differences in ESG adoption, regulatory alignment, and digital infrastructure. Methodologically, using real-time ESG data analytics, text mining of sustainability reports, or a hybrid model combining PLS-SEM and deep learning could improve the accuracy and reliability of predictive analytics. Lastly, deepening the paradigm to include the effects of ESG on broader measures of project success, such as operational efficiency, social value creation, and financial viability, would provide more information for academia and practitioners.

Author statement

We, the authors, declare that the manuscript entitled “*Integrating ESG Factors into Cost Forecasting for Sustainable Project Management: Empirical Evidence from Kazakhstan*” is original and has not been published previously, nor is it under consideration for publication elsewhere. Any prior dissemination in the form of a preprint, abstract, thesis, or academic presentation complies fully with the journal’s policies on prior publication. All authors have approved the manuscript and agree to its submission to *World Development Sustainability*. The work has been approved by the appropriate institutional authorities where the research was conducted. If accepted for publication, the manuscript will not be published elsewhere in the same form, in English or in any other language, without the written consent of the publisher. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI in scientific writing statement

During the preparation of this work, the authors used the Grammarly tool in order to minimize and produce a mistake-free manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Ethical statement

The authors affirm that this research was conducted in full compliance with the highest standards of academic integrity and ethical research practice. Human participants voluntarily responded to the survey after receiving and acknowledging an informed consent form outlining the study’s purpose and confidentiality assurances. No animals or sensitive personal data were involved, and institutional ethical approval was not required for this study. All secondary data sources have been appropriately acknowledged and cited. The authors declare no conflicts of interest, and all co-authors have reviewed and approved the final version of the manuscript for submission to *World Development Sustainability*.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. Due to institutional restrictions and privacy considerations involving student participants, the dataset is not publicly available.

CRedit authorship contribution statement

Meruyert Kussaiyn: Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rajasekhara Mouly Potluri:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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